



MOL-Mamba: Enhancing Molecular Representation with Structural & Electronic Insights

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Overview

- Background:** Molecular representation learning plays a crucial role in drug design.
- Challenges:** To accurately represent molecules, Graph Transformers GTs excel with self-supervised pretraining. But existing approaches often overlook the relationship between molecular structure and electronic information, resulting in incomplete molecular representation.
- Our Solutions:** To address this, we introduce MOL-Mamba framework combining structural and electronic insights.
 - An Atom & Fragment **Mamba-Graph (MG)** for hierarchical structural reasoning;
 - A **Mamba-Transformer (MT)** fuser for integrating molecular structure and electronic correlation learning.
- Objectives:** The goal is to learn graph and e-descriptor encoders which maps the input graphs and e-descriptors to a vector representation without any label information.
- Finetuning:** The learned encoders can then be used for various downstream property prediction tasks through finetuning.

Mol-Mamba with different pretraining loss settings:

$\mathcal{L}_d$	$\mathcal{L}_s$	$\mathcal{L}_f$	$\mathcal{L}_{mask}$	BBBP	Tox21	ESOL	Lipo
-	-	-	-	72.5 (±0.4)	78.6 (±0.6)	0.70 (±0.03)	0.63 (±0.03)
✓	-	-	-	73.4 (±0.3)	79.1 (±0.5)	0.68 (±0.02)	0.59 (±0.02)
✓	✓	-	-	73.8 (±0.3)	80.5 (±0.5)	0.66 (±0.02)	0.58 (±0.02)
✓	✓	✓	-	74.2 (±0.3)	80.9 (±0.4)	0.65 (±0.02)	0.56 (±0.02)
✓	✓	✓	✓	75.0 (±0.2)	81.3 (±0.4)	0.63 (±0.01)	0.53 (±0.01)

Main module ablation results:



Methods



- Pretraining Stage 1:** molecular structure learning with the Distribution Collaborative Training
- Pretraining Stage 2:** E-semantic Fusion Training for molecular structural & electronic fusion learning

Results

Results of state-of-arts on seven classification benchmarks:

	Method	Venue	BBBP 2,039	Tox21 7,831	ClinTox 1,478	HIV 41,127	BACE 1,513	SIDER 1,427	MUV 93,087
supervised	SchNet	NIPS'17	84.8 (±2.2)	77.2 (±2.3)	71.5 (±3.7)	70.2 (±3.4)	76.6 (±1.1)	53.9 (±3.7)	71.3 (±3.0)
	GIN	ICLR'19	65.8 (±4.5)	74.0 (±0.8)	58.0 (±4.4)	75.3 (±1.9)	70.1 (±5.4)	57.3 (±1.6)	71.8 (±2.5)
	AttentiveFP	JMC'19	64.3 (±1.8)	76.1 (±0.5)	84.7 (±0.3)	75.7 (±1.4)	78.4 (±2.2)	60.6 (±3.2)	76.6 (±1.5)
	DMPNN	NIPS'21	71.2 (±3.8)	68.9 (±1.3)	90.5 (±5.3)	75.0 (±2.1)	85.3 (±5.3)	63.2 (±2.3)	76.2 (±2.8)
	PretrainGNN	ICLR'20	68.7 (±1.3)	78.1 (±0.6)	72.6 (±1.5)	79.9 (±0.7)	84.5 (±0.7)	62.7 (±0.8)	81.3 (±2.1)
pretraining	GROVER	NIPS'20	69.5 (±0.1)	73.5 (±0.1)	76.2 (±3.7)	68.2 (±1.1)	81.0 (±1.4)	65.4 (±0.1)	67.3 (±1.8)
	GEM	NMI'22	72.4 (±0.4)	78.1 (±0.1)	90.1 (±1.3)	80.6 (±0.9)	85.6 (±1.1)	67.2 (±0.4)	81.7 (±0.5)
	GraphMVP	ICLR'22	72.4 (±1.6)	74.4 (±0.2)	77.5 (±4.2)	77.0 (±1.2)	81.2 (±0.9)	63.9 (±1.2)	75.0 (±1.0)
	MolCLR	NMI'22	73.6 (±0.5)	79.8 (±0.7)	93.2 (±1.7)	80.6 (±1.1)	89.0 (±0.3)	68.0 (±1.1)	88.6 (±2.2)
	Uni-Mol	ICLR'23	72.9 (±0.6)	79.6 (±0.5)	91.9 (±1.8)	80.8 (±0.3)	85.7 (±0.2)	65.9 (±1.3)	82.1 (±1.3)
	MOLEBLEND	ICLR'24	73.0 (±0.8)	77.8 (±0.8)	87.6 (±0.7)	79.0 (±0.8)	83.7 (±1.4)	64.9 (±0.3)	77.2 (±2.3)
	MOL-Mamba (Ours)		75.0 (±0.2)	81.3 (±0.4)	92.7 (±1.1)	81.6 (±0.5)	86.4 (±0.3)	68.3 (±0.9)	89.0 (±1.2)

Descriptors selection:

Type of Descriptors	Notation	Dimension	Number
E-state	$\mathcal{D}_S$	2	25
Molecular Property	$\mathcal{D}_M$	2	55
Quantum Chemical	$\mathcal{D}_Q$	2	7
Charge	$\mathcal{D}_C$	2	25

Model complexity comparison:

Model	#Params	FLOPs	Runtime(s/molecule)	ROC-AUC
GIN	1.63M	1.2M	0.001	74.0 (±0.8)
SchNet	1.86M	1.5M	0.0011	77.2 (±2.3)
Uni-Mol	47.1M	10M	0.0083	79.6 (±0.5)
MOLEBLEND	47.1M	11M	0.0091	77.8 (±0.8)
MOL-Mamba	6.98M	4M	0.0012	81.3 (±0.4)

T-SNE visualization of our pretraining (PT) methods:



Exemplary explanations for using MG and MT modules:



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